Anthara CarePath: Indian Outpatient Referral Workflow Review and MVP Assessment

"Review Anthara CarePath: 24h ideathon checklist for clinician-referred outpatient visits, tracking tasks, verified/unknown details and visit summaries. Compare Indian referral tools, gaps and duplication; recommend a pivot if needed. Define an MVP, exclusions and data, privacy, safety and adoption risks. Cite recent peer-reviewed/official Indian sources (titles, authors, years, journals, DOIs), marking evidence versus assumptions. Rank by need, novelty, feasibility, adoption and impact."
Generated by LeapSpace

Anthara CarePath at a Glance

Key Findings Table

Question	Evidence-based finding	Population / setting	Citations
Is there a real problem?	Yes. Indian referral pathways show non-compliance, delayed specialist presentation, incomplete pre-referral management, weak counselling, and tertiary crowding.	Eye care, obstetrics, rheumatology, infant infection care, public OPD pathways	[1] [2] [3] [4] [5] [6] [7] [8] [9]
Does CarePath risk duplication?	Yes if it tries to own registration, appointments, queueing, generic teleconsultation, or broad EHR functions, because Indian systems already cover much of that.	ABDM/ABHA stack, ORS-linked flows, Scan and Share, eSanjeevani, hospital EHR/HIS	[10] [11] [12] [13] [14] [15] [16] [17] [18]
Where is the unmet gap?	The least-served layer is the referral handoff itself: referral reason, verified vs unknown details, pending tasks, patient instructions, loop closure, and a return summary/back-referral note.	Clinician-referred outpatient visits across fragmented systems	[16] [17] [19] [20] [21] [22]

Question	Evidence-based finding	Population / setting	Citations
What form is most feasible?	A lightweight workflow tool is more feasible than a broad platform because adoption drops with dual entry, connectivity dependence, poor usability, and technician reliance.	Public and mixed-care digital implementations in India	[23] [24] [25] [26] [27] [28]
What should the MVP do?	Narrow MVP: structured referral intake, verified/unknown fields, pending-task checklist, patient instructions/red flags, one-episode status tracking, specialist summary, and return note.	One specialty pathway first	[21] [28] [29] [30]
What are the biggest risks?	Privacy, over-sharing, breach risk, dual documentation burden, weak GP integration, missing feedback loops, and workflow complexity.	Digital health exchange and tele-referral in India	[16] [17] [25] [31] [32] [33] [34] [35]

Direct Answer

Anthara CarePath addresses a genuine Indian outpatient problem, but it should **pivot from "broad referral platform" to "referral handoff assistant"** for clinician-referred visits [1] [4] [19] [20]. Evidence supports high need, but also shows that registration, appointments, tokening, teleconsultation, and some referral plumbing are increasingly covered by ABDM/ABHA, ORS-linked flows, Scan and Share, eSanjeevani, and existing hospital systems [10] [12] [14] [15] [16] [18].

Recommended product direction: build a thin layer that improves the quality and closure of one referral episode: what the referrer suspects, what is verified, what is unknown, what must happen before the visit, what the patient must do, and what the specialist sends back afterward [17] [21] [22].

Best MVP:

1. structured referral intake,
2. verified vs unknown tags,
3. pending-task checklist,
4. patient instructions and red flags,
5. one-screen specialist visit summary/back-referral,
6. minimal status tracking for one visit episode,

7. export/share as document attachment rather than full-stack interoperability on day 1 [24] [29] [30].

Do not build first: booking engine, queue management, teleconsult app, full EHR, AI diagnosis, payment, population analytics, or cross-hospital workflow orchestration [10] [16] [23] [28].

Rank of the pivoted concept: Need = High; Novelty = Medium; Feasibility = Medium-High if lightweight; Adoption = Medium; Impact = Medium-High in referral-heavy specialties [1] [5] [25] [29] [36].

Confidence level: Medium-High - strong evidence for the problem and adoption constraints, weaker direct Indian evidence on minimum outpatient handoff standards and head-to-head product duplication [1] [37] [38].

Study Scope

This review covers Indian evidence and official digital-health implementations from roughly 2018-2025 across outpatient referral pathways, telemedicine, OPD registration, EHR/EMR adoption, privacy/governance, and implementation science [7] [14] [31] [36]. It includes peer-reviewed studies, implementation reports, and policy-oriented analyses summarized in the supplied evidence base [18] [39].

Assumptions & Limitations

- The supplied evidence **does not include the full bibliographic metadata** requested in the query, so titles, authors, journals, years, and DOIs cannot be reproduced reliably here without fabricating details [meta_analisys].
- There is **no strong Indian benchmark** in the retrieved set for minimum outpatient referral-summary fields or direct comparisons against all HIS/ABDM modules, so some MVP design elements are reasoned assumptions rather than established standards [37] [38] [40].
- Claims about the exact best specialty wedge and export mode are **inferences**, not direct outcome evidence [17] [36].

Suggested Further Research

- Run a pilot comparing structured handoff sheets versus usual referral notes on completion, specialist preparedness, and repeat visits [21] [29].
- Test whether “verified/unknown” labeling improves safety and trust versus forcing all fields complete [25] [41].
- Compare duplication risk directly against ABDM-linked HIS workflows in one hospital network [12] [22].
- Study clinician willingness to complete structured summaries when dual documentation is minimized versus required [24] [25].
- Define a minimum Indian outpatient handoff dataset through multi-site consensus and workflow testing [30] [40].

Foundational context

A referral tool can target at least four distinct layers, and the evidence matters because these layers are already unevenly digitized in India [10] [17] [22]:

1. **Front door**: registration, identity, tokening, appointments [10] [42].
2. **Consult channel**: teleconsultation or virtual follow-up [15] [39].
3. **Clinical record**: EHR/EMR storage and retrieval [18] [43].
4. **Handoff layer**: why the patient is being referred, what is known, what is missing, what needs to happen before the visit, and what comes back after specialist review [16] [19] [21].

CarePath is most defensible only if it focuses on the **fourth layer** [16] [22].
Confidence level: High - this aligns with the strongest cross-source pattern on duplication versus unmet need [12] [17] [22].

Why the problem matters

Referral breakdown is well documented in Indian care pathways. Eye-care studies found referral non-compliance rates of 31.6%, 24.2%, and 37.8% in different cohorts, with economic and attitudinal barriers prominent [1] [2] [3]. Obstetric and rheumatology pathways show delayed advice and delayed specialist referral, including scan advice after 20 weeks in fetal anomaly pathways and rheumatology delays beyond two years in many patients [4] [5]. In young-infant infection care, many cases remained at primary level when hospital referral was not accepted, with delayed care-seeking and manpower/supply gaps complicating management [6].

Public-system evidence also shows weak gatekeeping. Patients often bypass lower levels and directly access tertiary services, crowding OPDs and increasing waits [8] [9] [44]. Referral paperwork exists but is often incomplete enough that documentation quality itself becomes part of the safety problem [21].

Taken together, these findings suggest that the need is real, but the failures are not purely informational: affordability, transport, awareness, counselling, and system design all matter [19] [20] [45]. A checklist product can improve handoff quality, but it cannot by itself fix referral compliance or access barriers [6] [7].

Confidence level: High - consistent findings across multiple specialties and settings [1] [4] [7].

How referrals are actually made in India

Real-world referral behavior is often informal. Physicians rely on specialist availability, accessibility, reputation, and personal networks because comprehensive provider databases are limited [19]. Referral counselling and caregiver-patient communication are often inadequate [19]. Broader analyses describe unstructured referral networks, weak monitoring, and information asymmetry [20]. Tele-referral pathways also show missing re-referrals and weak feedback loops, especially where GP integration is poor [16] [17].

This is the product opportunity: not discovering specialists, but **making the referral packet and return communication usable** [17] [19].

Confidence level: High - direct evidence supports informal behavior and weak communication [19] [20].

What already exists in India

Front-door tools

ABDM/ABHA provides interoperable identity and access infrastructure, including registries and pathways intended to support appointment and service access [10] [11] [13] [14]. Scan and Share and online OPD registration systems reduce waiting substantially through QR and token flows [42] [46]. Therefore, a CarePath product that starts with “book/register/check-in faster” would mostly duplicate active infrastructure [10] [42].

E-consult and tele-referral

Indian teleconsultation tools already support audio, video, text, prescriptions, continuity, and in some cases referral generation [15] [39] [47] [48] [49] [50]. But they are inconsistent in feedback loops and GP integration, and suffer from doctor availability, stock-outs, long waits, and connectivity problems [16] [17]. So the gap is not “teleconsult exists?” but “does the referral episode close properly?” [17].

EHR / interoperability capability

Indian EHR models emphasize coordination, summarization, and provider communication, and at least one cloud pilot reported cross-level sharing [18] [43]. But adoption barriers are severe: poor usability, funding gaps, weak training, network issues, and higher burden at high patient volumes [23] [24]. Simultaneous paper and digital recording creates dual workload and worsens consistency [24].

Duplication versus unmet need

Already covered elsewhere:

- patient identity and registries [10]
- booking/token/front-door queueing [42] [46]
- generic teleconsult modes [15] [39]

- broad EHR ambitions [18] [43]

Still weak or inconsistent:

- structured referral reason and urgency [21]
- verified versus unknown details [inference from documentation gaps and referral incompleteness: 1-2-3-6,3-16]
- pending-task visibility before visit [29]
- patient instructions/red flags attached to the referral [30]
- specialist return summary/back-referral loop [16] [17]

Confidence level: Medium-High - strong evidence on existing digital layers; weaker direct head-to-head evidence on duplication [37].

MVP design for CarePath

Recommended MVP workflow

Evidence-backed core:

1. Referral intake from clinician/facility [21].
2. Structured episode sheet with required and optional fields [30].
3. Mark each critical field as **verified**, **patient-reported**, or **unknown** [inference from incomplete referral/documentation evidence: 1-2-3-6,3-16].
4. Task checklist before visit: attach tests, confirm medicines, confirm allergies if known, give patient instructions [29] [30].
5. One-episode status tracking: referred / booked / seen / summary sent [assumption grounded in narrow workflow feasibility: 5-2-4-4,2-2-3-8].
6. Specialist visit summary and return note to referrer [17] [30].

This design matches the strongest opportunity: better documentation completeness and continuity within existing workflows, not system-wide referral reform [24] [29].

Minimum data elements

Evidence-supported candidate fields: history, findings, diagnostic tests, treatment plan, consent/referral papers, discharge or summary records [30]. Referral document quality work also supports capturing structured transfer details [21]. Structured chart-review literature supports fields such as allergy history, diagnosis/problem lists, medication records, follow-up actions, and consult details, though not from Indian outpatient studies [51] [52] [53].

Proposed MVP field set (assumption informed by evidence):

- patient identity/contact
- referring clinician/facility
- reason for referral
- urgency/red flags
- relevant diagnosis/history
- current medicines
- allergies
- prior tests attached/missing
- verified / unknown flags
- patient counselling given
- pending tasks
- specialist impression
- medication changes
- follow-up advice
- back-referral plan [21] [30] [51]

Exclusions and non-goals

Do **not** include in MVP:

- appointment engine or queue management [10] [42]
- teleconsult platform [15]
- full EHR replacement [18] [43]
- AI diagnosis or risk scoring [54]
- deep interoperability across all hospitals at launch [22]
- broad analytics/secondary-use dashboards [33]

Pivot triggers

Pivot harder toward a narrow workflow if:

- users must enter the same data twice [24] [25]
- use depends on technicians or constant connectivity [24] [25]
- average added visit time is more than trivial [26] [29]
- uptake is low outside one specialty [55] [56]
- hospitals ask for booking/queueing features rather than handoff quality [10] [46]

Confidence level: Medium - the narrowness recommendation is strongly supported, but exact workflow states and fields are partly design inference [28] [36].

Privacy, safety, and adoption risks

Privacy and consent

Indian scholarship consistently treats consent as necessary but insufficient. Health-data governance needs purpose limitation, stronger accountability, privacy-by-design, and safeguards against export, aggregation, re-identification, and secondary reuse [31] [32] [33] [57]. The DPDPA-era literature also highlights uncertainty for downstream reuse and the lack of a distinct sensitive-health-data category in some analyses [58] [59] [60].

MVP implication: share only minimum necessary data for care coordination, show explicit notice, support withdrawal/revocation where feasible, and block analytics reuse by default [31] [32].

Security and auditability

Centralized record stores increase breach risk, so role-based or purpose-based access control and auditable access logs are core requirements [34] [35] [61] [62] [63]. Blockchain-style architectures are discussed in the literature, but the evidence does not show they are necessary for this MVP [64] [65] [66]. A conventional auditable architecture is likely enough for a thin handoff layer [34] [35].

Adoption barriers

Dual data entry sharply increases workload [25]. Connectivity gaps, low awareness, training needs, and weak uptake remain common [25] [26] [67]. eSanjeevani-related evidence also shows poor GP integration and missing feedback loops [16] [17]. Therefore, adoption depends on making CarePath a **document-and-task layer over existing behavior**, not a new destination app [24].

Patient safety failure modes

Safety risk comes from false completeness, missing allergy/medication details, delayed escalation, and over-complex workflow [41] [68] [69]. The “verified/unknown” concept may reduce false certainty by making missingness explicit rather than hidden [inference: 1-2-3-6,3-16].

Confidence level: High for privacy and adoption risks; **Medium** for the specific safety value of “verified/unknown” because that is a design inference rather than directly tested Indian evidence [25] [33] [41].

Recommendation and prioritization

Comparative ranking

Option	Need	Novelty	Feasibility	Adoption	Impact	Overall
Broad outpatient app/platform	High	Low	Low	Low	Medium	Not recommended [10] [16] [23] [25]
Registration/queue tool	Medium	Low	High	High	Medium	Duplicative [42] [46]
Teleconsult/referral marketplace	Medium	Low-Medium	Medium	Medium	Medium	Crowded, feedback gaps remain [15] [17]
Referral handoff assistant	High	Medium	Medium-High	Medium	Medium-High	Best option [19] [22] [25] [36]

Recommended product direction

Anthara CarePath should **launch only as a narrow referral handoff assistant**, ideally in one specialty with recurrent referral loss or documentation friction [1] [5] [28]. It should plug into existing hospital or ABDM-linked workflows where possible, and prove value through better completeness, faster specialist preparation, and loop closure [12] [22].

Evidence versus assumptions

Directly supported by evidence

- Referral handoff problems are real and common [1] [4] [19].
- Front-door and telehealth layers already exist in India [10] [15].
- Complex digital workflows face adoption barriers [24] [25] [26].
- Privacy and audit controls are essential [32] [34].

Assumptions / inferred design recommendations

- “Verified vs unknown” should be a signature differentiator [21] [41].
- One-episode specialist return-summary flow is the best initial wedge [17] [36].
- The best first deployment is in a referral-heavy specialty rather than general OPD [1] [5].

Bottom line

Anthara CarePath should **pivot, not stop**. The evidence supports the problem, but not a broad platform play [1] [22]. The most defensible MVP is a **lightweight referral handoff assistant** that improves documentation quality, clarifies unknowns, tracks a few pre-visit tasks, and closes the loop with a specialist summary [17] [21] [29].

Final confidence level: Medium-High - strong on need, duplication risk, privacy/adoption constraints, and the case for narrow scope; weaker on exact minimum field standards because the supplied Indian evidence does not define them directly [37] [40].

References

1. Khanna, R.C., Kim, S., Giridhar, P., (...), Rao, G.N. (2018) Barriers to uptake of referral services from secondary care to tertiary care and its associated factors in L v Prasad Eye Institute network in Southern India: A cross-sectional study. *BMJ Open*. https://doi.org/10.1136/bmjopen-2017-020687
2. Padhy, D., Pyda, G., Marmamula, S., Khanna, R.C. (2024) Barriers to uptake of referral services from secondary eye care to tertiary eye care and its associated determinants in L V Prasad Eye Institute network in Southern India: A crosssectional study-Report II. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0303401
3. Padhy, D., Pyda, G., Khanna, R.C. (2025) Barriers to referral uptake from primary to secondary eye care in the L V Prasad Eye Institute Network in South India. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0325974
4. Yadav, M., Gowda, M., Chinta, N., (...), Murugesan, S. (2023) Factors associated with late gestational age of diagnosis and/or delayed referral of fetuses with major structural malformations: A study in a tertiary care hospital in South India. *Prenatal Diagnosis*. https://doi.org/10.1002/pd.6431
5. Chandrashekara, S., Shenoy, P., Kumar, U., (...), Sharma, B.L. (2024) Blockades in the Pathway of Specialty Care in Rheumatoid Arthritis: A Report Based on Indian Rheumatology Association Database. *Journal of Association of Physicians of India*. https://doi.org/10.59556/japi.72.0495
6. Goyal, N., Rongsen-Chandola, T., Sood, M., (...), Bhandari, N. (2020) Management of possible serious bacterial infection in young infants closer to home when referral is not feasible: Lessons from implementation research in Himachal Pradesh, India. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0243724
7. Fatma, N., Kala, K., Ramamohan, V. (2025) Towards mitigating overcrowding in urban Indian healthcare facilities: stakeholder perception analysis for existing and potential new patient referral mechanisms. *BMC Health Services Research*. https://doi.org/10.1186/s12913-025-13602-y
8. Ugargol, A.P., Mukherji, A., Tiwari, R. (2023) In search of a fix to the primary health care chasm in India: can institutionalizing a public health cadre and inducting family physicians be the answer?. *The Lancet Regional Health - Southeast Asia*. https://doi.org/10.1016/j.lansea.2023.100197
9. Olickal, J.J., Chinnakali, P., Suryanarayana, B.S., (...), Subrahmanyam, D.K.S. (2023) Down referral and assessing comprehensive diabetes care in primary care settings: An operational research from India. *Diabetes and Metabolic Syndrome: Clinical Research and Reviews*. https://doi.org/10.1016/j.dsx.2022.102694
10. Kamath, Rajesh, Banu, Muneera, Shet, Nagaraj, (...), Kamath, Sagarika (2025) Awareness of and Challenges in Utilizing the Ayushman Bharat Digital Mission for Healthcare Delivery: Qualitative Insights from University Students in Coastal Karnataka in India. *Healthcare*. https://doi.org/10.3390/healthcare13040382
11. https://www.sciencedirect.com/science/article/pii/S1936645025001524
12. https://www.sciencedirect.com/science/article/pii/S3005077424000568

13. Shah, Priyansh (2023) International telehealth implementation—the Indian experience, project CardioGram. *Emerging Practices in Telehealth*. https://doi.org/10.1016/B978-0-443-15980-0.00008-9

14. Malhotra, Shefali (2023) Cyberattacks hold up India's push for digitisation of health. *The BMJ*. https://doi.org/10.1136/bmj.p263

15. Dinakaran, D., Manjunatha, N., Kumar, C.N., Math, S.B. (2021) Telemedicine practice guidelines of India, 2020: Implications and challenges. *Indian Journal of Psychiatry*. https://doi.org/10.4103/psychiatry.IndianJPsychiatry_476_20

16. Dastidar, B.G., Jani, A.R., Suri, S., Nagaraja, V.H. (2024) Reimagining India's National Telemedicine Service to improve access to care. *The Lancet Regional Health - Southeast Asia*. https://doi.org/10.1016/j.lansea.2024.100480

17. Kotwani, P., Pandya, A., Patwardhan, V., (...), Patel, G.M. (2025) System-Level Enablers and Barriers To Telemedicine Adoption: Perspectives of Community Health Officers Delivering Primary Care in India. *SN Comprehensive Clinical Medicine*. https://doi.org/10.1007/s42399-025-02220-8

18. Ganiga, R., Pai, R.M., Pai, M.M., Sinha, R.K. (2020) A preliminary study of real-time capturing and sharing of routine health data among the public health professionals. *Indian Journal of Community Medicine*. https://doi.org/10.4103/ijcm.IJCM_258_19

19. Sagayam, M.S., Gupta, P., Ramesh, R., Sengupta, A. (2025) Urban Indian healthcare referral system: A qualitative exploration from the physicians' perspectives. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0338113

20. Sagayam, M.S., Sengupta, A., Mazundar, S. (2025) Healthcare Referral, Market Imperfection and Health Policies in India: A Populist Argument. *Journal of Health Management*. https://doi.org/10.1177/09720634251326492

21. Saini, A., Agarwal, M., Singh, V.K., (...), Jauhari, S. (2023) Completeness and legibility of obstetric referral documents from peripheral public health facilities to a tertiary care facility in North India- a cross-sectional study. *Clinical Epidemiology and Global Health*. https://doi.org/10.1016/j.cegh.2023.101342

22. Pai, M.M.M., Ganiga, R., Pai, R.M., Sinha, R.K. (2021) Standard electronic health record (EHR) framework for Indian healthcare system. *Health Services and Outcomes Research Methodology*. https://doi.org/10.1007/s10742-020-00238-0

23. Gururaj, H.L., Soundarya, S.B.C., Basavesha, D. (2023) Medical Record Management for Disease Management. *Intelligent Systems in Healthcare and Disease Identification using Data Science*. https://doi.org/10.1201/9781003354178-6

24. Shilpa, D.M., Naik, P., Shewade, H., Sudarshan, H. (2020) Assessing the implementation of a mobile App-based electronic health record: A mixed-method study from South India. *Journal of Education and Health Promotion*. https://doi.org/10.4103/jehp.jehp_749_19

25. Jat, M., Gupta, M., Prinja, S. (2025) Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India: Cross-Sectional Study. *JMIR Formative Research*. https://doi.org/10.2196/69824

26. Tyagi, M., Kumar, D., Sharma, K., (...), Awasthi, S. (2025) Barriers and enablers of using a clinical decision support algorithm to consult sick children at primary health facilities: A qualitative study from Uttar Pradesh, India. *Clinical Epidemiology and Global Health*. https://doi.org/10.1016/j.cegh.2025.102038

27. Vedachalam, S.K., Siromany, V.A., VanderEnde, D., (...), Mathur, P. (2024) Implementing a healthcare-associated bloodstream infection surveillance network in India: a mixed-methods study on the best practices, challenges and opportunities, 2022. *Antimicrobial Resistance and Infection Control*. https://doi.org/10.1186/s13756-024-01501-6

28. Faujdar, D.S., Singh, T., Kaur, M., (...), Kumar, R. (2021) Stakeholders' perceptions of the implementation of a patient-centric digital health application for primary healthcare in india. *Healthcare Informatics Research*. https://doi.org/10.4258/HIR.2021.27.4.315

29. Srilatha, B., Jeba, J.J., Susithra, D.N., (...), Joseph, L. (2024) Palliative care outpatients and improved documentation—what matters most? Quality improvement project. *BMJ Supportive and Palliative Care*. https://doi.org/10.1136/spcare-2023-004650

30. Thomas, J. (2009) Medical records and issues in negligence. *Indian Journal of Urology*. https://doi.org/10.4103/0970-1591.56208

31. Christopher, D., Mundhra, J., Pradhan, N., Kabra, A. (2025) India's Data Protection Regime Notified: Overview of the Act and Rules. *Global Privacy Law Review*. https://doi.org/10.54648/gplr2025023

32. Saksena, N., Matthan, R., Bhan, A., Balsari, S. (2021) Rebooting consent in the digital age: A governance framework for health data exchange. *BMJ Global Health*. https://doi.org/10.1136/bmjgh-2021-005057

33. Shrivastava, S., Srikanth, T.K., Vs, D. (2020) E-Governance for healthcare service delivery in India: Challenges and opportunities in security and privacy. *ACM International Conference Proceeding Series*. https://doi.org/10.1145/3428502.3428527

34. Yakubu, Bello Musa, Ali, Syeda Mahera, Khan, Majid Iqbal, Bhattarakosol, Pattarasinee (2024) PatCen: A blockchain-based patient-centric mechanism for the granular access control of infectious disease-related test records. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0310407

35. Wong, Brian Kee Mun, Vengusamy, Sivakumar, Wee, Gabriel Wei En (2025) Digital healthcare business ecosystems in the Asia-Pacific: Transformative dynamics and strategic imperatives. *Digital Healthcare, Digital Transformation and Citizen Empowerment in Asia-Pacific and Europe for a Healthier Society*. https://doi.org/10.1016/B978-0-443-30168-1.00006-2

36. Proctor, E., Silmere, H., Raghavan, R., (...), Hensley, M. (2011) Outcomes for implementation research: Conceptual distinctions, measurement challenges, and research agenda. *Administration and Policy in Mental Health and Mental Health Services Research*. https://doi.org/10.1007/s10488-010-0319-7

37. Ved, R.R., Gupta, G., Singh, S. (2019) India's health and wellness centres: realizing universal health coverage through comprehensive primary health care. *WHO South-East Asia journal of public health*. https://doi.org/10.4103/2224-3151.255344

38. Kleczka, Bernadette, Kumar, Pratap, Njeru, Mercy Karimi, (...), Marx, Michael (2019) Using rubber stamps and mobile phones to help understand and change antibiotic prescribing behaviour in private sector primary healthcare clinics in Kenya. *BMJ Global Health*. https://doi.org/10.1136/bmjgh-2019-001422

39. Nagaraja, V.H., Ghosh Dastidar, B., Suri, S., Jani, A.R. (2024) Perspectives and use of telemedicine by doctors in India: A cross-sectional study. *Health Policy and Technology*. https://doi.org/10.1016/j.hlpt.2024.100845

40. Koli, P.G., Kshirsagar, N.A., Shetty, Y.C., (...), Parmar, U. (2019) A systematic review of standard treatment guidelines in India. *Indian Journal of Medical Research*. https://doi.org/10.4103/ijmr.IJMR_902_17

41. Das, P., Biswas, N., Saha, N., (...), Gupta, A. (2024) PRESCRIPTION AUDIT AT A TERTIARY CARE HOSPITAL - A CROSS-SECTIONAL STUDY. *Asian Journal of Pharmaceutical and Clinical Research*. https://doi.org/10.22159/ajpcr.2024v17i11.52531

42. B Neogi, S.B., Pagaria, V., Dutta, S., (...), Samtani, R. (2025) Scan and share to register outpatients, India. *Bulletin of the World Health Organization*. https://doi.org/10.2471/BLT.24.292828

43. Moholkar, K., Sayyad, A., Mogle, S., (...), Jayabhaye, S. (2024) Comprehensive Analysis of Electronic Health Records: Introducing SWASTHA - A Novel Model. *2nd International Conference on Sustainable Computing and Smart Systems, ICSCSS 2024 - Proceedings*. https://doi.org/10.1109/ICSCSS60660.2024.10625324

44. Grover, Ashoo, Bhargava, Balram, Srivastava, Saumya, (...), Kabra, Sushil Kumar (2023) Developing Standard Treatment Workflows—way to universal healthcare in India. *Frontiers in Public Health*. https://doi.org/10.3389/fpubh.2023.1178160

45. Jindal, Devraj, Sharma, Hanspria, Gupta, Yashdeep, (...), Tandon, Nikhil (2022) Improving care for hypertension and diabetes in india by addition of clinical decision support system and task shifting in the national NCD program: I-TREC model of care. *BMC Health Services Research*. https://doi.org/10.1186/s12913-022-08025-y

46. Sharma, N., Aggarwal, A.K., Arora, P., Bahuguna, P. (2022) Association of waiting time and satisfaction level of patients with online registration system in a tertiary level medical institute outpatient department (OPD). *Health Policy and Technology*. https://doi.org/10.1016/j.hlpt.2022.100687

47. Jain, S., Arora, S., Chauhan, A., (...), Singh, M. (2024) Super-Specialty Medical Teleconsultations to Remote Areas of Ladakh, India, via Esanjeevani Under the Hub-and-Spoke Model: A Pilot Study. *Telehealth and Medicine Today*. https://doi.org/10.30953/thmt.v9.471

48. Basavarajappa, C., Sandeep, S., Dalal, P., (...), Math, S.B. (2022) Current telepsychiatry practice in India - An online survey of psychiatrists. *Indian Journal of Psychiatry*. https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_331_21

49. Sujithra Devi, R., Naik Parvathi, T., Rupesh Bala Murugan, R., (...), Priyadarshini, T. (2022) Outcomes of teleconsultation services and patient satisfaction among pregnant women delivering at a tertiary care center in South India during coronavirus disease 2019 pandemic. *Indian Journal of Public Health*. https://doi.org/10.4103/ijph.ijph_1942_21

50. Das, A.V., Rani, P.K., Vaddavalli, P.K. (2020) Tele-consultations and electronic medical records driven remote patient care: Responding to the COVID-19 lockdown in India. *Indian Journal of Ophthalmology*. https://doi.org/10.4103/ijo.IJO_1089_20

51. Clark, S., Boggs, K.M., Balekian, D.S., (...), Cephas, J. (2019) Changes in Emergency Department Concordance with Guidelines for the Management of Food-Induced Anaphylaxis: 1999-2001 versus 2013-2015. *Journal of Allergy and Clinical Immunology: In Practice*. https://doi.org/10.1016/j.jaip.2019.04.004

52. Patel, Anita K., Olson, Taylor, Ray, Christopher, (...), Pollack, Murray M. (2025) Clinical assessment of the criticality index – dynamic, a machine learning prediction model of future care needs in pediatric inpatients. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0320586

53. Cruvinel, E., Ellerbeck, E., Richter, K.P., (...), Scheuermann, T.S. (2023) Hospital to Outpatient Transitions of Care for Tobacco Treatment. *Chest*. https://doi.org/10.1016/j.chest.2023.01.028

54. Batko, K., Ślezak, A. (2022) The use of Big Data Analytics in healthcare. *Journal of Big Data*. https://doi.org/10.1186/s40537-021-00553-4

55. Khanna, A., Kumar, A., Rajeshwari, B.S. (2025) Advancing Digital Health: Strategies for Implementation and Management. *Proceedings of the 3rd International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics, IITCEE 2025*. https://doi.org/10.1109/IITCEE64140.2025.10915449

56. Aravind, A.P., Soundappan, K. (2024) Perception towards electronic health records & uptake of digital health IDs among the urban residents in northern India: A mixed methods study. *Indian Journal of Medical Research*. https://doi.org/10.25259/ijmr_664_23

57. Kasyapa, M.S.B., Vanmathi, C. (2023) A Privacy Protection Mechanism for Indian Health Care Data Exchange Using Blockchain. *2023 Innovations in Power and Advanced Computing Technologies, i-PACT 2023*. https://doi.org/10.1109/I-PACT58649.2023.10434770

58. Kolanu, M., Lakra, R., Shrivastava, A. (2025) Data, Control, and Power: Decoding India's Digital Personal Data Protection Act, 2023. *Global Privacy Law Review*. https://doi.org/10.54648/gplr2026002

59. Singh, K.K. (2024) India's Legal and Policy Approach to Personal Data Protection and Cross-Border Data Transfer: A Critical Study with Special Reference to Health Data. *Perspectives in Law, Business and Innovation*. https://doi.org/10.1007/978-981-97-9983-1_11

60. Parsheera, S. (2025) AWAITING THE DAWN OF DATA PROTECTION REGULATION IN INDIA. *Personal Data Architectures in the BRICS Countries*. https://doi.org/10.1093/9780198974468.001.0004

61. https://www.sciencedirect.com/science/article/pii/S1936645025000804

62. Chen, L., Feng, T., Ma, R., Shi, J. (2024) BTMDS: Blockchain trusted medical data sharing scheme with privacy protection and access control. *Computer Communications*. https://doi.org/10.1016/j.comcom.2024.07.007

63. https://www.sciencedirect.com/science/article/pii/S1573048425009021

64. Ahmad, Mohammad, De Alwis, Chamitha, Shukla, Mitul, Sant, Paul (2024) Privacy-preserving patient-centric electronic health records exchange using blockchain. *Artificial Intelligence, Big Data, Blockchain and 5G for the Digital Transformation of the Healthcare Industry*. https://doi.org/10.1016/B978-0-443-21598-8.00020-8

65. Qu, X., Yang, Z., Chen, Z., Sun, G. (2025) A consent-aware electronic medical records sharing method based on blockchain. *Computer Standards and Interfaces*. https://doi.org/10.1016/j.csi.2024.103902

66. https://www.sciencedirect.com/science/article/pii/S0219137724001036

67. Parameshwarappa, P.M., Olickal, J.J. (2023) Telemedicine Awareness and the Preferred Digital Healthcare Tools: A Community-based Cross-sectional Study from Rural Karnataka, India. *Indian Journal of Community Medicine*. https://doi.org/10.4103/ijcm.ijcm_770_22

68. Almathami, H.K.Y., Win, K., Vlahu- Gjorgievska, E. (2020) Barriers and facilitators that influence telemedicine-based, real-time, online consultation at patients' homes: Systematic literature review. *Journal of Medical Internet Research*. https://doi.org/10.2196/16407

69. Cowan, K.E., McKean, A.J., Gentry, M.T., Hilty, D.M. (2019) Barriers to Use of Telepsychiatry: Clinicians as Gatekeepers. *Mayo Clinic Proceedings*. https://doi.org/10.1016/j.mayocp.2019.04.018